

Terry Nyirenda

AI ENGINEER

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PROFESSIONAL SUMMARY

AI Engineer building production-style multi-agent systems and backend infrastructure. Experienced designing FastAPI services, LangGraph workflows, CrewAI agents, PostgreSQL and Redis-backed architectures, automated testing, CI/CD and LLM integrations. 2+ years evaluating frontier language models at Invisible Technologies combined with independent development of AI systems that solve real business problems.

PROFESSIONAL HIGHLIGHTS

- Independently built three end-to-end AI automation platforms integrating LLMs, REST APIs, workflow orchestration, and backend services.
- 2+ years evaluating and improving frontier language models at Invisible Technologies.
- Combines AI integration, backend engineering, and business process automation across independent projects.
- Master's degree in Business Mathematics and Informatics — a strong quantitative foundation for AI system design.

TECHNICAL SKILLS

Programming Languages

- Python
- C++
- SQL

Backend

- FastAPI
- REST APIs
- Pydantic

AI Engineering

- LangGraph
- CrewAI
- OpenAI API
- Prompt Engineering
- LLM Evaluation
- Claude Code
- OpenCode

Infrastructure

- PostgreSQL
- Redis
- Docker Compose
- GitHub Actions
- SQLAlchemy

Authentication

- OAuth2
- JWT

Automation & Orchestration

- n8n
- Workflow Automation
- Airtable
- Slack Integrations
- Email Automation

Development & Testing

- Git / GitHub
- Automated Testing (Pytest)
- Error Handling & Retry Logic
- Postman

Data & Analysis

- Excel
- Statistical Analysis
- Data Quality Assurance

PROJECTS

Independent projects, built and documented outside of employment.

SupportOps AI

Production-style multi-agent customer support platform

github.com/Terrytd0/Supportops-AI
Demo Video

- Designed and built a multi-agent customer support platform using LangGraph, CrewAI, FastAPI, PostgreSQL, Redis and Docker Compose.
- Designed a policy-gated workflow that classifies customer requests, routes them to specialist agents, and escalates high-risk cases for supervisor approval.
- Implemented JWT authentication, Redis-backed rate limiting, audit logging, idempotent request handling, and knowledge-base retrieval.
- Designed reusable specialist agent abstractions enabling Billing, Technical, Account, and General support agents to share common tooling and orchestration.
- Wrote an extensive automated test suite covering API endpoints, graph execution, policies, repositories and integration workflows.
- Implemented a GitHub Actions CI pipeline that automatically runs linting, type checking and automated tests on every push and pull request.

Technologies: Python, FastAPI, LangGraph, CrewAI, PostgreSQL, Redis, Docker Compose, GitHub Actions, JWT, SQLAlchemy, Pytest, Ruff, mypy

Finance Intelligence Platform

Independent Project — End-to-end AI-powered financial data automation platform

- Designed and built a FastAPI backend exposing REST endpoints for financial data ingestion and AI-driven analysis.
- Built a seven-stage automated pipeline — ingestion, validation, KPI calculation, AI-driven anomaly detection, executive reporting, notifications, and audit logging.
- Integrated the OpenAI API to power anomaly detection and generate executive-ready business reports.
- Orchestrated n8n workflow automation, built automated tests, retry logic, and technical documentation.

Technologies: Python, FastAPI, OpenAI API, n8n, Airtable, Slack API, Pytest, Claude Code

github.com/Terrytd0/Finance-Intelligence-Platform

Demo Video

Lead Intelligence Automation Platform

Independent Project — Automated sales-intelligence and lead-enrichment workflow

- Built an automated pipeline that eliminates manual lead research using OpenAI-powered company enrichment.
- Orchestrated the end-to-end workflow with n8n, connecting Google Forms intake (via Apps Script and a Cloudflare Tunnel webhook) to AI processing and CRM updates.
- Integrated Airtable as a lightweight CRM with duplicate-detection and returning-contact logic, plus AI guardrails that return structured fallback responses for invalid or unmatched input.

Technologies: OpenAI API, n8n, Airtable, Google Apps Script, Cloudflare Tunnel, Slack API, Gmail SMTP

github.com/Terrytd0/AI-Lead-Intelligence-Platform

PROFESSIONAL EXPERIENCE

Advanced AI Data Trainer

Invisible Technologies

- Collaborate on evaluation of frontier language models using structured prompting, reasoning analysis and quality assessment across complex real-world tasks.
- Conducted systematic error analysis on AI-generated outputs, identifying and documenting failure patterns to drive model quality improvements.
- Applied prompt engineering techniques to simulate realistic, complex user scenarios and expand evaluation coverage of edge cases.

May 2024 – Present

Remote

BMI Intern

Absa Group

- Performed quantitative analysis and predictive modeling to support stress testing and scenario analysis for enterprise risk assessment.
- Conducted research and statistical analysis supporting model development, validation, and business risk decision-making.
- Contributed to predictive modeling initiatives within structured, data-driven business analytics workflows.

Aug 2022 – Dec 2022

Johannesburg, South Africa

CERTIFICATIONS

IBM Full Stack Software Developer Professional Certificate

Salesforce Certified Administrator

Jun 2025

May 2026

EDUCATION

Master's Degree, Business Mathematics and Informatics

North-West University

- Jan 2022 – Mar 2023
- Potchefstroom, South Africa

Honours Degree, Quantitative Risk Management

North-West University

- Jan 2021 – Dec 2021
- Potchefstroom, South Africa

Bachelor of Science, Actuarial Science

North-West University

- Jan 2018 – Dec 2020
- Potchefstroom, South Africa